THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 10

Full Syllabus (Chapters 1–14) | Class 10 Mathematics | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 36	$2^2 \times 3^2$
2	**(a)** Student A is correct	It fails for three numbers in general
3	**(c)** 23	$8 + 15$
4	**(b)** Incorrect — it has exactly one zero	$ax + b = 0 \Rightarrow x = -\frac{b}{a}$
5	**(c)** infinitely many	$\frac{2}{4} = \frac{3}{6} = \frac{9}{18}$
6	**(c)** $k \neq 6$	$\frac{k}{3} \neq \frac{2}{1} \Rightarrow k \neq 6$
7	**(c)** $3, 4$	$(x-3)(x-4)$
8	**(d)** 1	$k - 3 + 2 = 0$
9	**(d)** 40	$49 - 3 \times 3 = 40$
10	**(b)** 30	$12, \ldots, 99$: $\frac{99-12}{3} + 1 = 30$
11	**(a)** Correct	Ratio of sides 1, equal angles
12	**(a)** SSS	All three sides proportional
13	**(a)** 2	$\frac{1}{\cos 60^\circ}$
14	**(c)** 80°	$180^\circ - 100^\circ$
15	**(d)** 30°	$\frac{360^\circ}{12}$
16	**(b)** 120°	$20 \times 6^\circ$
17	**(c)** 1 : 3	$\frac{1}{3}\pi r^2 h : \pi r^2 h$
18	**(b)** $l + \frac{\frac{n}{2} - cf}{f} \times h$	Standard formula
19	**(c)** Both get the same mean	Mean does not depend on the choice of a
20	**(b)** $\frac{1}{3}$	$\{3, 5\}$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	All three are primes, so HCF $= 1$ and LCM $= 17 \times 23 \times 29 = 11339$. For distinct primes, LCM = their product.	1+1
22	$x^2 - 4x + 4 = (x-2)^2 \Rightarrow$ both zeroes equal 2; the graph only touches the x-axis at $(2, 0)$. Claim is **false**.	1+1
23	$D = 36 - 24 = 12 > 0 \Rightarrow$ two distinct real roots	1+1
24	$\frac{PE}{EQ} = \frac{4}{4.5} = \frac{8}{9}$, $\frac{PF}{FR} = \frac{8}{9}$ — equal $\Rightarrow EF \parallel QR$ (converse of BPT)	1+1
25	$\sqrt{(2a)^2 + (2b)^2} = 2\sqrt{a^2 + b^2}$	1+1
26	In right $\triangle ABC$, $\sin^2 A + \cos^2 A = \frac{BC^2 + AB^2}{AC^2} = 1$ by Pythagoras — true for every acute angle (e.g. $\theta = 30^\circ$: $\frac{1}{4} + \frac{3}{4} = 1$).	1+1
27	Height above his eyes $= 20 \tan 45^\circ = 20$ m; tower $= 21.8$ m. Claim is **false** (eye height ignored).	1+1
28	In quadrilateral OPTQ, $\angle OPT = \angle OQT = 90^\circ$; angle sum $360^\circ \Rightarrow \angle POQ + \angle PTQ = 180^\circ$	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Both pass through $(1, 0), (0, -1)$ — coincident $\Rightarrow$ consistent, infinitely many solutions	2+1
30	$\Sigma fx = 15 + 105 + 375 + 350 + 225 = 1070$; mean $= \frac{1070}{40} = 26.75$	1+1+1
31	Middle angle $= 60^\circ$ (sum 180°) $\Rightarrow d = 20^\circ \Rightarrow 40^\circ, 60^\circ, 80^\circ$	1+1+1
32	$2x + 9 = x + 4 + 7 \Rightarrow x = 2$; $AB = 4$, $BC = 6$ cm	1+1+1
33	$19.625 - 12.5 = 7.125$ cm²	1+1+1
34	(i) $\frac{6}{36} = \frac{1}{6}$ (ii) $\frac{3}{36} = \frac{1}{12}$ (iii) $\frac{1}{6}$	1+1+1
35	Distances $\frac{100}{\sqrt{3}} \approx 57.74$ m and 100 m; moved 42.26 m in 2 min $\Rightarrow \approx 21.13$ m/min	1+1+1
	OR: $60\sqrt{3} + 20\sqrt{3} = 80\sqrt{3} \approx 138.56$ m	1+1+1

36	$n = 40$; cf 6, 15, 25 ⇒ $40 + \frac{20-15}{10} \times 20 = 50$; the student is wrong	1+1+1
	OR: $a = 60, h = 20$; $\Sigma fu = -16 - 12 + 0 + 6 + 8 = -14$ ⇒ $60 - \frac{280}{50} = 54.4$	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) Green $(2, 25)$, red $(8, 20)$ (ii) $\sqrt{36 + 25} = \sqrt{61}$ m (iii) mid-point $(5, 22.5)$: on the 5th line, 22.5 m along	2+2+1
	OR: (i) Each side $= \sqrt{18} = 3\sqrt{2}$ (ii) $AC = BD = 6$ (iii) Equal sides and equal diagonals ⇒ square; **Champa** is correct	2+2+1
38	$m^2 - n^2 = 4\tan\theta\sin\theta$; $mn = \tan^2\theta - \sin^2\theta = \sin^2\theta\left(\frac{1}{\cos^2\theta} - 1\right) = \sin^2\theta\tan^2\theta$ ⇒ $4\sqrt{mn} = 4\sin\theta\tan\theta$ ✓	2+2+1
	OR: (i) $\sqrt{144 + 25} = 13$ m (ii) $\frac{5}{13}, \frac{12}{13}, \frac{5}{12}$ (iii) $\frac{25+144}{169} = 1$ ✓ (iv) False: $\tan\theta = \frac{5}{12} < 1$ (rise < base)	1+2+1+1
39	Cylinder 2 m. (i) $12.32 + 5.749 = 18.069$ m³ ≈ 18069 L (ii) $2\pi rh + 2\pi r^2 = 17.6 + 12.32 = 29.92$ m²	3+2
	OR: Height $2r$ ⇒ cylinder height r. $\pi r^3 + \frac{2}{3}\pi r^3 = \frac{5}{3}\pi r^3 = \frac{880}{21} \Rightarrow r^3 = 8$, $r = 2$ ⇒ height $= 4$ m	2+2+1
40	$x + y = 24$; median class 500–600: $525 = 500 + \frac{50-(36+x)}{20} \times 100 \Rightarrow x = 9, y = 15$	2+2+1
	OR: Median: cf 2, 10, 22 ⇒ $150 + \frac{25-22}{20} \times 10 = 151.5$ cm. Mean: $a = 145, h = 10$, $\Sigma fu = 24$ ⇒ 149.8 cm	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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